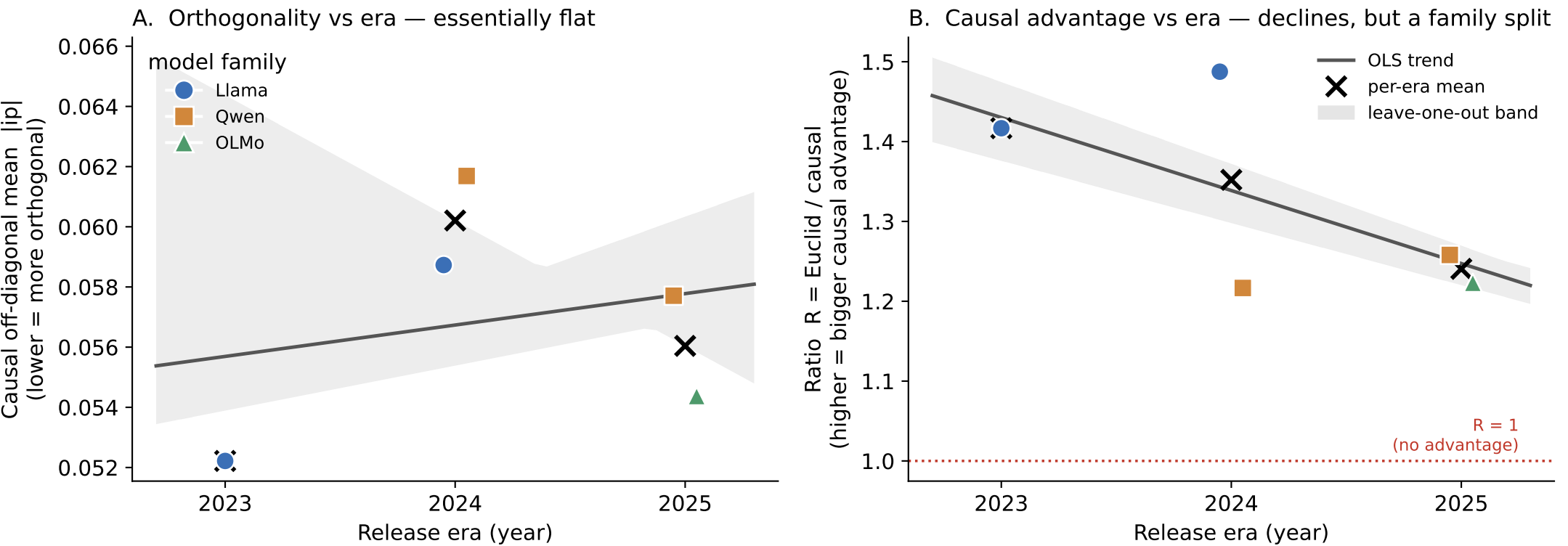


Study 002 — Causal-inner-product geometry across the 2023-2025 base-model ladder (5 models, 22-concept intersection)



5 data points $\Rightarrow$ descriptive fit, not a scaling law. Causal off-diagonal is flat across eras (slope +0.001/yr, $R^2=0.06$); the ratio R declines with year ($R^2=0.38$) but the LOO band and family markers show the decline is a Llama $\rightarrow$ Qwen/OLMo family split (R changes $\sim 4.8\times$ more across families than within a family step), not a scale effect (R vs log-params corr ≈ -0.09). $R > 1$ on all 5.